Received: 23 January 2020 | Revised: 20 April 2020 | Accepted: 6 May 2020
DOI: 10.1002/dta.2818

RESEARCH ARTICLE

WILEY

Searching for new long-term urinary metabolites of metenolone and drostanolone using gas chromatography–mass spectrometry with a focus on non-hydrolysed sulfates

Aðalheiður Dóra Albertsdóttir[1] | Wim Van Gansbeke[1] | Gilles Coppieters[1] | Kyzylkul Balgimbekova[2] | Peter Van Eenoo[1] | Michael Polet[1]

[1]Doping Control Laboratory, Department of Diagnostic Sciences, Ghent University, Zwijnaarde, Belgium

[2]The Athletes' Anti-Doping Laboratory, Committee for Sport and Physical Education, Ministry of Culture and Sport of the Republic of Kazakhstan, Almaty, Kazakhstan

Correspondence
Aðalheiður Dóra Albertsdóttir, Doping Control Laboratory, Department of Diagnostic Sciences, Ghent University, Technologiepark 30 B, B-9052 Zwijnaarde, Belgium.
Email: aalheiurdora.albertsdottir@ugent.be

Funding information
Partnership for Clean Competition, Grant/Award Number: 2019R2000409G

Abstract

Sulfated metabolites have been shown to have potential as long-term markers of anabolic–androgenic steroid (AAS) abuse. In 2019, the compatibility of gas chromatography–mass spectrometry (GC–MS) with non-hydrolysed sulfated steroids was demonstrated, and this approach allowed the incorporation of these compounds in a broad GC–MS initial testing procedure at a later stage. However, research is needed to identify which are beneficial.

In this study, a search for new long-term metabolites of two popular AAS, metenolone and drostanolone, was undertaken through two excretion studies each. The excretion samples were analysed using GC–chemical ionization–triple quadrupole MS (GC–CI–MS/MS) after the application of three separate sample preparation methodologies (i.e. hydrolysis with *Escherichia coli*–derived β-glucuronidase, *Helix pomatia*–derived β-glucuronidase/arylsulfatase and non-hydrolysed sulfated steroids).

For metenolone, a non-hydrolysed sulfated metabolite, 1β-methyl-5α-androstan-17-one-3ζ-sulfate, was documented for the first time to provide the longest detection time of up to 17 days. This metabolite increased the detection time by nearly a factor of 2 in comparison with the currently monitored markers for metenolone in a routine doping control initial testing procedure. In the second excretion study, it prolonged the detection window by 25%.

In the case of drostanolone, the non-hydrolysed sulfated metabolite with the longest detection time was the sulfated analogue of the main drostanolone metabolite (3α-hydroxy-2α-methyl-5α-androstan-17-one) with a detection time of up to 24 days. However, the currently monitored main drostanolone metabolite in routine doping control, after hydrolysis of the glucuronide with *E.coli*, remained superior in detection time (i.e. up to 29 days).

KEYWORDS
drostanolone, GC–MS, metenolone, steroids, sulfates

1 | INTRODUCTION

The majority of adverse analytical findings (AAFs) in sport doping control in the past years involved anabolic–androgenic steroid (AAS) misuse.[1–7] Furthermore, the advancement of instrumentation and the identification of new AAS long-term metabolites have been credited with over 100 additional AAFs after the reanalysis of samples from the 2008 Beijing and 2012 London Olympic games.[8] The high frequency of detection of AASs points to their continuous popularity among professional athletes and underlines the importance of the perpetual search for new long-term metabolites to both catch the transgressors and deter their use. Sulfates are phase II–type metabolites and are promising candidates in that endeavour as the discovery of sulfate metabolites of mesterolone,[9] metandienone,[10] methyltestosterone[11] and clostebol[12] has resulted in extended detection times. In light of this, a search for new long-term metabolites of metenolone (17β-hydroxy-1-methyl-5α-androst-1-en-3-one) and drostanolone (17β-hydroxy-2α-methyl-5α-androstan-3-one) was undertaken as these two steroids have consistently been listed among the top 10 of the most frequently detected AASs in the World Anti-doping Agency testing figures report over the past years.[1–7]

Metenolone (Figure 1) is a synthetic AAS used for the treatment of anaemia caused by bone marrow failure.[13] Its anabolic properties exceed those of testosterone[14] and is orally active despite not having a methyl group in the 17α position, which results in lower hepatoxicity.[13,15] Therefore, it is not surprising that it is a frequently detected anabolic agent.[1–7] Metenolone undergoes extensive metabolism in humans, and its misuse is monitored mainly by the parent compound and its major metabolite 3α-hydroxy-1-methylene-5α-androstan-17-one using gas chromatography–mass spectrometry (GC–MS) after the hydrolysis of the glucuronides with *Escherichia coli*–derived β-glucuronidase.[16] The two most recent studies, Fragkaki et al.[17] and He et al.,[18] employed liquid chromatography–high-resolution mass spectrometry and liquid chromatography quadrupole time of flight (LC-QToF), respectively, to investigate its metabolism. Fragkaki's search, through two excretion studies by oral administration, yielded eight sulfates of which the four most important ones were identified as the sulfate analogues of metenolone, 3α-hydroxy-1-methylene-5α-androstan-17-one, 3ξ-hydroxy-1β-methyl-5α-androstan-17-one and 16β-hydroxy-1-methyl-5α-androst-1-ene-3,-17-dione. In this study, sulfates and the hydrolysed glucuronides had comparable detection windows of approximately 12 days.[17] He et al.[18] identified 16 metabolites in total through a single excretion study by intravenous injection. Contrary to Fragkaki, sulfates provided superior detection over glucuronides with the most promising one being 1-methylene-5α-androstan-3,17-dion-2ξ–sulfate (detectable for 40 days). In the study of He et al., the sulfated analogues of the metabolites mentioned by Fragkaki still had almost double the detection time of the glucuronides.

Drostanolone (Figure 2) has secondary antiestrogenic activity[19] and does not cause a significant virilizing effect.[20] Therefore, its therapeutic use includes treatment of breast cancer in women as a propionate ester.[21] It is one of the most frequently detected AASs,[1–7]

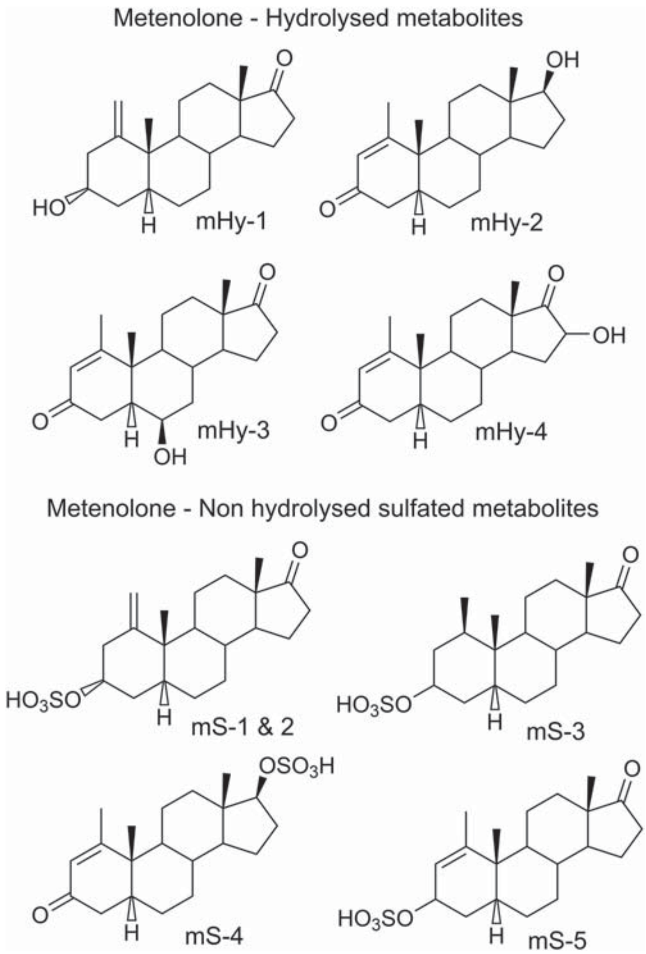
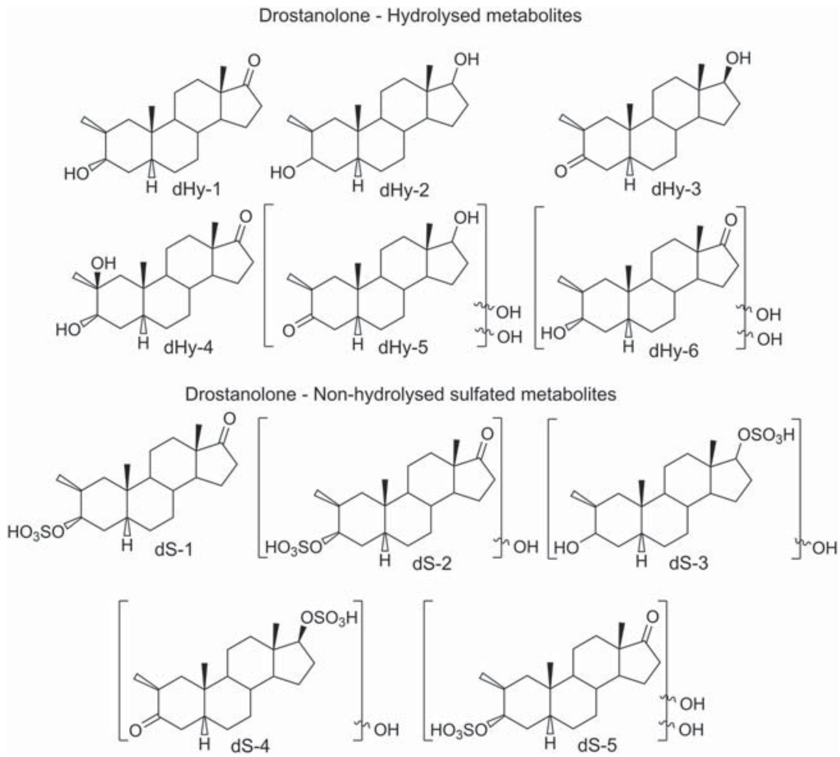


FIGURE 1 Metenolone (mHy-2) and its metabolites

and its misuse is monitored by GC–MS focusing on the 3α-hydroxy-2α-methyl-5α-androstan-17-one metabolite after the hydrolysis of the glucuronide with *E. coli*–derived β-glucuronidase.[22] The metabolite was identified, among others, in a metabolism study conducted on drostanolone using rabbits and GC–MS back in the 1970s, and it was concluded to be the major metabolite.[23] De Boer et al.[24] also concluded it to be a major metabolite in humans when analysing samples collected from athletes who were self-administering drostanolone; furthermore, it was established that the metabolites were conjugated to glucuronic acid. Liu et al.,[25] by analysing a single excretion study with intravenous administration on an LC-QToF, reported five sulfated, five glucuronidated and one free metabolites. The previously reported glucuronidated metabolite proved to have the longest detection time, (29 days), whereas its sulfated analogue was present for 14 days. The sulfated metabolite, which proved to have the longest detection time (i.e. 24 days), was identified as 2β-hydroxy-2α-methyl-5α-androstan-17-one-3α-sulfate.[25]

Due to requirements of cost-effectiveness and high throughput in anti-doping laboratories, the general set-up of the initial screening consists of two complementary methods, one on GC–MS and the other on LC–MS, capable of detecting over 300 doping substances of various low- to medium-molecular-weight drug classes.[26–28] Despite

FIGURE 2 Drostanolone (dHy-3) and its metabolites



the aforementioned prolonged detection times resulting from targeting AAS sulfated metabolites on LC–MS, at this stage and to the best of our knowledge, sulfated steroid metabolites are not included in doping control screening methods. Analysing sulfated AAS on LC–MS requires a specific sample preparation that is incompatible with the wide range of compounds that need to be detected. Furthermore, the currently applied LC–MS initial testing procedures in most laboratories rely on high-resolution acquisition instrumentation, whereas LC triple quadrupole MS is the instrument of choice for the detection of sulfated steroids.

Therefore, for most AASs, the standard detection protocol remains GC triple quadrupole MS (GC–MS/MS) analysis after the hydrolysis of the glucuronides with *E. coli*–derived β-glucuronidase.[29] Recently, the compatibility of GC–MS with non-hydrolysed sulfated steroids was demonstrated, and, in the future, this approach will most likely allow the incorporation of these types of compounds in a broad GC–MS initial testing procedure.[30] Their chromatographic and mass spectrometric fragmentation behaviour was studied, and the collected data showed that the sulfate group was cleaved off in the injection port of the GC–MS, forming two isomers. Furthermore, new sulfate metabolites of mesterolone were discovered, which were previously not reported by LC–MS. The detection times on both GC–MS and LC–MS were comparable.

To be able to include non-hydrolysed sulfated AAS in a broad GC–MS initial testing procedure, research needs to identify which non-hydrolysed sulfated metabolites are beneficial (i.e. detection window extension) for inclusion. Due to substantial differences between GC–MS and LC–MS, one cannot assume that a (long-term) metabolite will have a comparable sensitivity and detection time with both techniques.[17,31] As such, in this study, an approach is set up to identify and characterize new (non-hydrolysed sulfated) metabolites on GC–MS for metenolone and drostanolone. The investigation involved three different sample preparation procedures, that is hydrolysis with *E. coli*–derived β-glucuronidase, *Helix pomatia*–derived β-glucuronidase/arylsulfatase and non-hydrolysed sulfated steroids.

2 | MATERIAL AND METHODS

2.1 | Reagents and reference standards

For the excretion studies, Primobolan® (metenolone acetate) was obtained from Schering (Berlin, Germany) and drostanolone propionate from UCB (Brussels, Belgium). The following reference standards were obtained from the National Measurement Institute (Pymble, Australia): drostanolone (17β-hydroxy-2α-methyl-5α-androstan-3-one), main drostanolone metabolite (3α-hydroxy-2α-methyl-5α-androstan-17-one), metenolone metabolite (3α-hydroxy-1-methylene-5α-androstan-17-one) and mesterolone metabolite (3α-hydroxy-1α-methyl-5α-androstan-17-one). Metenolone (17β-hydroxy-1-methyl-5α-androst-1-en-3-one) was a gift from the Drug Control Centre of King's College London (London, UK). The following salts were purchased from Merck (Darmstads, Germany): sodium chloride (NaCl), sodium sulfate (Na_2SO_4), potassium carbonate (K_2CO_3), disodium hydrogen phosphate (Na_2HPO_4), sodium dihydrogen phosphate ($NaH_2PO_4 \cdot 2H_2O$) and sodium acetate (NaOAc). Sodium hydrogen carbonate ($NaHCO_3$) was obtained from Fisher Scientific (Loughborough, UK), ammonium iodide (NH_4I) and saturated ammonium hydroxide (NH_4OH) was purchased from Merck (Darmstadt). β-Glucuronidase derived from *E. coli K12* (*E. coli*) and β-glucuronidase/arylsulfatase derived from *H. pomatia* were obtained from Roche Diagnostics GmbH (Mannheim, Germany). Ethyl acetate (EtAc) was obtained from Acros Organics (Geel, Belgium), methanol (MeOH) was purchased from Fisher Scientific and methyl *tert*-butyl ether (MTBE) was obtained from Biosolve (Valkenswaard, The Netherlands). *N*-Methyl-*N*-(trimethylsilyl) trifluoroacetamide (MSTFA) was obtained from Karl Bucher Chemische Fabrik GmbH (Waldstetten, Germany), ethanethiol was obtained from Acros. *N*,*N*-Dimethylformamide (DMF), 1,4-dioxane and sulfur trioxide pyridine complex (SO_3·py) (98%) were obtained from Sigma (St. Louis, MO, USA), and acetic acid was obtained from Sigma-Aldrich (Bornem, Belgium). Formic acid (FA) was purchased from Avantor (Deventer, The Netherlands).

The phosphate buffer (pH 7) was prepared by dissolving 7.1 g of $Na_2HPO_4 \cdot 2H_2O$ and 1.4 g of $NaH_2PO_4 \cdot H_2O$ in 100 mL of water. The acetate buffer (pH 5.2) was prepared by dissolving 136 g of NaOAc in 900 mL of water, and acetic acid was added until a pH of 5.2 was reached. The carbonate buffer (pH 9.5) was obtained by dissolving 135 g K_2CO_3 and 111 g $NaHCO_3$ in 900 mL of double-distilled water.

2.2 | Excretion studies

For metenolone acetate, two volunteers received orally a single 50 mg dose. For the first excretion study (healthy male, 36 years, 65 kg), blank urine was collected before administration (0 h), and afterwards samples were collected at 3, 5, 8 h and then every 12 h from day 1 to day 23. For the second one (healthy male, 32 years, 75 kg), blank urine was collected before administration and then every 24 h from day 1 to day 51.

In the case of drostanolone propionate, a single 25 mg dose was administered orally to two volunteers. For the first excretion study (healthy male, 31 years, 80 kg), a pre-administration sample was collected, then after 3, 5, 8 and 12 h, then every 12 h until day 5 and subsequently every 24 h until day 42. For the second one (healthy male, 36 years, 65 kg), a pre-administration urine sample was collected and samples at 1, 3, 5, 8 and 12 h and subsequently every 12 h until day 5 were collected. Then, samples were gathered every 24 h until day 29.

In all four instances, the administration studies were conducted according to the Helsinki Declaration, and written consent was acquired from all volunteers regarding the use of samples for research purposes (Ethics Committee Kazakh National Medical University and Ethics Committee University Hospital Ghent, EC/UZG2008/293).

The storage condition for all urine samples until analysis was −20°C.

2.3 | Sample preparation

The sample preparation for the hydrolysis of urine samples with *E. coli*–derived β-glucuronidase (i.e. hydrolysis of glucuronides) was performed according to a previously reported method[29] except 2 mL of urine was used instead of 1 mL. In short, 1 mL of phosphate buffer and 50 μL of β-glucuronidase derived from *E. coli* were added to 2 mL of urine. After incubation (i.e. 1.0 h at 56°C), 1 mL of carbonate buffer (pH 9.5) and 5 mL of MTBE were added and the tubes rolled for 20 min. The organic layer was transferred and evaporated to dryness under nitrogen atmosphere.

The sample preparation for the hydrolysis using *H. pomatia*–derived β-glucuronidase/arylsulfatase (i.e. hydrolysis of sulfates and glucuronides) was performed in the same manner except that an acetate buffer was used instead of phosphate buffer and the incubation was extended to 1.5 h.

The non-hydrolysed sulfates were extracted using a previously reported method.[30] In short, to 2 mL of urine, 1 mL of carbonate buffer (pH 9.5), 200 mg of NaCl and 3 mL of EtAc were added. The tubes were rolled for 20 min, the organic phase was transferred and, after adding another 3 mL of EtAc to the original tube, it was rolled for an additional 20 min. After transferring and combining the organic layers, they were evaporated to dryness under nitrogen atmosphere.

The derivatization procedure was the same in all three cases: 60 μL of MSTFA/ethanethiol/NH_4I (500:4:2, v/v/v) was added to the dried residue. After brief vortexing (2 s), the liquid was transferred to a vial, incubated at 80°C for 30 min and 2 μL injected on the GC–MS.

For the analysis on LC–MS/MS, the evaporated residue was dissolved and transferred as described by Fragkaki et al.[17] in 100 μL mixture of 80:20 ratio of the aqueous and organic mobile phases and 5 μL injected on the LC–MS/MS.

2.4 | Synthesis of sulfated steroids

The synthesis followed a previously published protocol.[32] In short, to a dried residue of the free steroid standard, 100 μL of 1,4-dioxane was added and briefly vortexed. Then, 100 μL of freshly prepared sulfur trioxide pyridine complex mixture was added, the resulting solution was stirred for 4 h and then 1.5 mL of water was added to stop the reaction. Subsequently, the mixture was loaded onto an Oasis Wax 3 cc cartridge (60 mg, 30 μm) that had been preconditioned with 5 mL of methanol and 5 mL of water. Washing was performed with 5 mL of 2% (v/v) formic acid in water, 5 mL of water and 5 mL of methanol. The sulfated steroids were eluted with 5 mL of 5% (v/v) saturated ammonia solution in methanol, the organic solvent was evaporated and the same derivatization procedure as in Section 2.3 was applied.

2.5 | Instrumentation

2.5.1 | Gas chromatography-chemical ionization-triple quadrupole mass spectrometry

For the chemical ionization measurements, an Agilent GC 7890 gas chromatograph coupled to an Agilent 7010B triple quadrupole mass spectrometer (QQQMS, Agilent Technologies, Palo Alto, CA, USA) was used, and ammonia was used as a reagent gas at a flow of 20%

(i.e. standard instrument setting). The sample was injected using the high-end PAL3 sample injector (CTC analytics, Switzerland). The inlet was operated at 275°C in splitless mode.

The GC was equipped with a 12 m × 250 μm I.D., 0.25 μm film thickness HP-1MS column (Agilent). The run time of the method was 14.761 min, and the starting temperature of the oven was 110°C (0.1 min). The temperature gradient was as follows: 70°C/min to 125°C (0.1 min) → 35°C/min to 186°C (0.05 min) → 2.2°C/min to 204°C → 20°C/min to 245°C → 45°C/min to 270°C → 75°C/min to 320°C (1.1 min). The carrier gas was helium with a constant flow rate of 0.9 mL/min until 9.2 min and then increased from 1.5 to 4 mL/min. For the mass spectrometer, nitrogen was used as collision gas with a flow rate of 1.5 mL/min, and helium was used as a quenching gas at a flow rate of 2.25 mL/min. The used modes were product ion scan (PIS) and multiple reaction monitoring (MRM).

2.5.2 | Gas chromatography-low energy electron ionization-quadrupole time of flight-mass spectrometry

The electron ionization (EI) runs were performed using an Agilent 7,250 GC–QToF–MS (Agilent Technologies) equipped with a 15 m × 200 μm I.D., 0.11 μm film thickness Ultra 1 column (Agilent Technologies). The sample (0.8 μL) was injected using an Agilent 7693A ALS, and the inlet was in splitless mode at 250°C. The oven was initially set to 110°C (0.1 min), but the temperature gradient was as follows: 75°C/min to 125°C (0.15 min) → 25°C/min to 165°C → 40°C/min to 185°C → 3.5°C/min to 210°C → 15°C/min to 240°C → 75°C/min to 325°C (0.8 min). The MS was operated in low energy electron ionization (LE-EI) mode at 17 eV, source temperature of 200°C and full scan high resolution from *m/z* 50 to 750.

2.5.3 | Liquid chromatography-tandem mass spectrometry

The LC–MS/MS runs were performed on a Dionex Ultimate 3,000 and an XRS open autosampler connected to a TSQ Vantage (both from Thermo Fisher Scientific). The type of column and mobile phases used, as well as the settings for the LC and MS/MS conditions, were adapted from Fragkaki et al.[17]

2.6 | Detection protocol

This procedure is based on a previously described methodology.[33] It uses positive chemical ionization (CI) with ammonia as a reagent gas in combination with GC triple quadrupole MS (GC–CI–MS/MS). This has proven to be more beneficial than EI as it is prone to less extensive fragmentation, offering the possibility of choosing product ions that have a higher mass to charge (*m/z*) ratio and thus are more selective leading to higher sensitivity.[29] In short, the process involves listing theoretically possible metabolites, predicting their CI transitions based on previously discovered correlations between AAS structure and CI-MS behaviour[34] and constructing a GC–CI–MS/MS MRM method using the predicted transitions. Then, excretion samples are analysed with the MRM method and metabolites identified. The metabolites with prolonged detection times are investigated further by optimizing the transitions and collision energies and analysing 20 blank urine samples with the MRM method to ensure their exogenous origin. Finally, one or multiple interesting metabolites are selected to be fully characterized. These metabolites have added value over the currently known and employed metabolites (i.e. longer detection times) and can be included in a routine screening method (i.e. initial testing procedure).

This methodology is applied to hydrolysed glucuronides (i.e. hydrolysis with β-glucuronidase derived from *E. coli*), hydrolysed sulfates (i.e. hydrolysis with β-glucuronidase/arylsulfatase derived from *H. pomatia*) and non-hydrolysed sulfates.

In the past, sulfates were generally hydrolysed with, for example *H. pomatia*–derived β-glucuronidase/arylsulfatase, prior to analysis on GC–MS. However, a recent study presented the chromatographic behaviour of non-hydrolysed sulfated steroids on both GC–CI–MS/MS and GC–LE–EI–QToF–MS, where, in contrast to LC–MS analysis, the intact derivatized sulfates were never detected. Instead, the sulfate group was cleaved off in the inlet, and overall the dominant species was demonstrated to be $[MH - H_2SO_4]^+$ in CI-MS/MS and $[M - H_2SO_4]^+$ in LE-EI-QToF.[30] As previously mentioned, the end goal is to incorporate the new long-term metabolites in a screening method. As doping control laboratories use GC–MS/MS to detect AAS use, it is necessary to evaluate which metabolites exhibit compatibility to GC–MS/MS to further advance the detection times of AAS as part of the continuous effort to combat doping in sport.

3 | RESULTS AND DISCUSSION

3.1 | Searching for metabolites

Possible metabolic pathways were identified using previously published articles investigating the metabolism of the steroids. For metenolone, known metabolic pathways include double bond reduction with and without 1-methylene formation, keto reduction and hydroxyl oxidation,[17,18,35] whereas the pathways for drostanolone are hydroxyl oxidation and keto reduction.[22,25,35] For both AASs, the addition of 1 or 2 hydroxyl groups was considered. The resulting theoretical metabolites are presented in Figure S1 (supporting information) for metenolone and Figure S2 (supporting information) for drostanolone. In total, 36 theoretical metabolites were considered for metenolone and 12 for drostanolone. However, if a metabolite did not contain a hydroxyl group, its consideration was omitted for the sulfated steroids.

Using a previously published work,[34] product ions were identified for each structure and a GC–CI–MS/MS MRM method was constructed for both metenolone and drostanolone. The precursor ions and product ions of the hydrolysed glucuronides and sulfates and non-hydrolysed sulfates for metenolone and drostanolone are listed in Table 1 and 2, respectively. A pre-administration (0 h) and two post-administration (8 h and 24 h) samples were extracted using the sample preparation described in Section 2.3 and analysed using the theoretical MRM method. The chromatograms were compared, possible metabolites identified and their retention time (Rt), precursor and product ions noted.

Then, product ion scans, using a collision energy of 10 and 20 eV, were carried out on the precursor ions of the detected metabolites to identify the most suitable product ions, preferably with higher intensity and specificity, and to use for characterization at later stages. This new information was used to construct a second optimized MRM method, which was applied to the analysis of excretion samples up to day 10. The metabolites with detection time under 7 days were excluded. To ensure that the remaining metabolites had an exogenous origin, 20 blank urine samples were run using the optimized method. In addition, for the non-hydrolysed sulfated metabolites, to exclude the remaining sulfates from being free metabolites, the post-administration sample was extracted again. This time the test tube containing the residue after evaporation of EtAc was washed with 2 mL of MTBE. The free metabolites would wash out with the MTBE, whereas the sulfated metabolites would not. Any residue of MTBE was evaporated before derivatization. Finally, all excretion samples were analysed to determine the detection window of the remaining metabolites.

To characterize the remaining metabolites, the excretion samples were run on GC–LE-EI-QToF–MS in full scan (FS) high resolution.

3.2 | Characterization of metenolone metabolites

After the elimination process, there were nine metabolites, four hydrolysed (mHy) and five non-hydrolysed sulfated (mS), remaining for metenolone (Figure 3). Table 3 lists the precursors and transitions of the remaining metabolites, and the accurate mass measurements are presented in Table S1 (supporting information).

3.2.1 | Hydrolysed metabolites

At this stage, routine doping control analyses involved the hydrolysis of the glucuronides with *E. coli*–derived β-glucuronidase. Only mHy-4 was detected longer with *H. pomatia*–derived β-glucuronidase/arylsulfatase, and thus it is concluded that it is longer excreted as a sulfate. The other three metabolites had longer or equal detection times with *E. coli*, meaning that these are excreted as glucuronides.

3.2.1.1 | mHy-1 and mHy-2

Both shared the same mass of *m/z* 447 ($[MH]^+$) in CI and *m/z* 446.3031 ($[M]^+$, $C_{26}H_{46}O_2Si_2$) in LE-EI-FS. By analysing their CI-PIS [Figure S3 (supporting information)] and comparing with the reference standard of metenolone and metenolone metabolite (3α-hydroxy-1-methylene-5α-androstan-17-one), we found that mHy-1 was metenolone metabolite and mHy-2 was the parent compound metenolone.

3.2.1.2 | mHy-3 and mHy-4

Both share the same mass of *m/z* 533 ($[MH]^+$) in CI [Figure S4 (supporting information)] and *m/z* 532.3219 ($[M]^+$, $C_{29}H_{52}O_3Si_3$) in LE-EI-FS. Compared to metenolone, mHy-3 and mHy-4 contain an additional hydroxyl and either a double bond formation or a 3,17-dione. As the formation of a double bond has not been reported before, it is more likely that the two metabolites are 3,17-dions. The LE-EI-FS spectra [Figure S5 (supporting information)] of mHy-4 have high abundance of *m/z* 517.2984 $[M - CH_3]^+$ (measured *m/z* 517.2984, $C_{28}H_{49}O_3Si_3$, error 0 ppm) and low abundance of *m/z* 195.1200 (measured *m/z* 195.1201, $C_{11}H_{19}OSi$, error 0.68 ppm), which Goudreault et al.[36] identify as characteristic of 16ζ-hydroxy-1-methyl-5α-androst-1-ene-3,17-dione. Furthermore, the LE-EI-FS spectra [Figure S5 (supporting information)] of mHy-3 matches that of 6β-hydroxy-1-methyl-5α-androst-1-ene-3,17-dione also in the work of Goudreault et al.[36] However, the synthesis of reference material is required to unequivocally confirm the definitive structures of mHy-3 and mHy-4.

3.2.2 | Non-hydrolysed sulfated metabolites

3.2.2.1 | mS-1, mS-2, mS-4 and mS-5

The four metabolites share the same mass of *m/z* 357 ($[MH - H_2SO_4]^+$) in CI [Figure S3 (supporting information)] and *m/z* 356.2530 ($[M - H_2SO_4]^+$, $C_{23}H_{36}OSi$) in LE-EI-FS. The synthesis of the sulfated metenolone metabolite (3α-hydroxy-1-methylene-5α-androstan-17-one) revealed that both mS-1 and mS-2 stemmed from that compound as both the CI spectra and retention time matched. The two peaks originated from the isomer formation that was induced when the sulfate was cleaved off in the inlet.[30] mS-4 is the sulfate form of metenolone as was confirmed by analysing the synthesized sulfated reference material. Most likely, mS-5 is a sulfated metenolone-like metabolite where the keto group is on C17 and the sulfate on C3. However, synthesis is required to support this claim.

3.2.2.2 | mS-3

The mass of mS-3 in CI was *m/z* 359 ($[MH - H_2SO_4]^+$) and *m/z* 358.2686 ($[M - H_2SO_4]^+$, $C_{23}H_{38}OSi$) in LE-EI-FS which points to the absence of a double bond. This leaves the possibility of 3-keto, 17-hydroxy or the reverse. As neither was available in-house, mesterolone metabolite (3α-hydroxy-1α-methyl-5α-androstan-17-one) was sulfated and analysed to aid in this endeavour. The CI-PIS of both mS-3 (Figure 4) and mesterolone metabolite contained the fragment ions *m/z* 147, 161 and 213, which are common of 3-hydroxy, 17-keto AAS.[34] This metabolite has been reported by Fragkaki et al.,[17] but, compared to mesterolone metabolite, with the C1 methyl group in

TABLE 1 CI MRM transitions derived from the structures of theoretical possible metabolites of metenolone

Theoretical metabolites	Precursors for hydrolysed metabolites [*m/z*]	Precursors for non-hydrolysed sulfated metabolites [*m/z*]	Product ions [*m/z*]
Meth I	447	357	267, 241, 209, 187, 181, 147
Meth II	449	359	269, 243, 187, 159, 157
Meth III	449	359	269, 243, 209, 201, 187, 149, 119
Meth IV	451	361	105, 93, 81
Meth V	447	357	267, 241, 209, 201, 187, 149, 119
Meth VI	449	359	187, 161, 149, 135, 119, 105
Meth VII	535	445	355, 329, 209, 201, 187,149, 119
Meth VIII	537	447	357, 331, 187, 157, 149
Meth IX	537	447	357, 331, 209, 187, 149, 119
Meth X	537	447	359, 333, 105, 93, 81
Meth XI	535	445	355, 329, 209, 201, 187, 149, 119
Meth XII	537	447	357, 331, 209, 201, 187, 149, 119
Meth XIII	623	533	443, 417, 209, 201, 187, 149, 119
Meth XIV	625	535	445, 419, 187, 159, 157
Meth XV	625	535	445, 419, 209, 201, 187, 149, 119
Meth XVI	627	537	447, 421, 105, 93, 81
Meth XVII	623	533	443, 417, 209, 201, 187, 149, 119
Meth XVIII	625	535	445, 419, 209, 201, 187, 149, 119
Meth XIX	445		355, 329, 209, 201, 187, 149, 119
Meth XX	447		357, 331, 159, 157
Meth XXI	447	357	267, 241, 209, 201, 187, 149, 119
Meth XXII	449	359	213, 199, 161, 159, 147, 145
Meth XXIII	445		355, 329, 209, 201, 187, 149, 119
Meth XXIV	447	357	267, 241, 209, 187, 149, 119
Meth XXV	533	443	353, 327, 209, 201, 187, 149, 119
Meth XXVI	535	445	355, 329, 159, 157
Meth XXVII	535	445	355, 241, 209, 201, 187, 149, 119
Meth XXVIII	537	447	357, 331, 213, 199, 161, 159, 147, 145
Meth XXIX	533	443	353, 327, 209, 201, 187, 149, 119
Meth XXX	535	445	355, 329, 209, 201, 187, 149, 119
Meth XXXI	621	531	441, 415, 209, 201, 187, 149, 119
Meth XXXII	623	533	443, 417, 187, 159, 157
Meth XXXIII	623	533	443, 241, 209, 201, 187, 149, 119
Meth XXXIV	625	535	445, 419, 213, 199, 161, 159, 147, 145
Meth XXXV	621	531	441, 415, 209,201, 187, 149, 119
Meth XXXVI	623	533	443, 417, 209, 201, 187, 149, 119

Note. The collision energy for all transitions was 30 eV.
Abbreviation: CI MRM, chemical ionization multiple reaction monitoring.

the β-position instead of the α-position. As a result, it is highly likely that the structure of mS-3 is 1β-methyl-5α-androstan-17-one-3ζ-sulfate, but the synthesis of reference material is required to shed light on the α/β orientation of the sulfate group. Despite the previous discovery of this metabolite, the novelty demonstrated in this paper is that this is the first time that this metabolite is identified as the longest detectable metabolite.

3.3 | Characterization of drostanolone metabolites

For drostanolone, there were 11 metabolites [6 hydrolysed (dHy) and 5 non-hydrolysed sulfated (dS)], remaining after the elimination process (Figure 5). Table 4 lists the precursors and transitions of the remaining metabolites, whereas in Table S2 (supporting information), the results of the accurate mass measurements are presented.

TABLE 2 CI MRM transitions derived from the structures of theoretical possible metabolites of drostanolone

Theoretical metabolites	Precursors for hydrolysed metabolites [*m/z*]	Precursors for non-hydrolysed sulfated metabolites [*m/z*]	Product ions [*m/z*]
Drost I	449	359	269, 229, 187, 173, 157
Drost II	449	359	269, 213, 173, 161, 145
Drost III	447		267, 187, 173, 157
Drost IV	451		361, 271, 245, 121, 107, 95
	361	361	271, 245, 121, 107, 95
	271	271	121, 107, 95
Drost V	537		447, 357, 267, 187, 173, 157
	447	447	357, 267, 187, 173, 157
		357	267, 187, 173, 157
Drost VI	537		447, 357, 267, 213, 173, 161, 145
	447	447	357, 267, 213, 173, 161, 145
		357	267, 213, 173, 161, 145
Drost VII	535	446	355, 265, 187, 173, 157
Drost VIII	539		449, 359, 269, 121, 107, 95
	449	449	359, 269, 121, 107, 95
	359	359	269, 121, 107, 95
	269	269	121, 107, 95
Drost IX	625		535, 445, 355, 265, 187, 173, 157
	535	535	445, 355, 265, 187, 173, 157
		445	355, 265, 187, 173, 157
		355	265, 187, 173, 157
Drost X	625	535	213, 173, 161, 145
		445	213, 173, 161, 145
		355	213, 173, 161, 145
Drost XI	623		533, 443, 353, 263, 187, 173, 157
	533	533	443, 353, 263, 187, 173, 157
Drost XII	627		537, 447, 121, 107, 95
	537	537	447, 121, 107, 95
	447	447	121, 107, 95
		357	121, 107, 95

Note. The collision energy for all transitions was 30 eV.
Abbreviation: CI MRM, chemical ionization multiple reaction monitoring.

3.3.1 | Hydrolysed metabolites

All metabolites except dHy-5 were longer detectable with *E. coli*–derived β-glucuronidase and thus are excreted as glucuronides. The metabolite dHy-5 was found only with *H. pomatia*–derived β-glucuronidase/arylsulfatase, and therefore it is excreted solely as a sulfate.

3.3.1.1 | dHy-1 and dHy-3

Both shared a mass of *m/z* 449 ($[MH]^+$) in CI [Figure S7 (supporting information)] and *m/z* 448.3187 ($[M]^+$, $C_{26}H_{48}O_2Si_2$) in LE-EI-FS. The CI-PIS of dHy-1 contained *m/z* 145, 161, 199 and 213, which are common fragments of 3-hydroxy-17-keto steroids.[34] This indicates that dHy-1 is 3α-hydroxy-2α-methyl-5α-androstan-17-one (main drostanolone metabolite), and it was verified by analysing the reference material of the main drostanolone metabolite. The CI-PIS spectra of dHy-3 contained $[MH - 44]^+$ (*m/z* 405), $[MH - 144]^+$ (*m/z* 305), $[MH - 144–90]^+$ (*m/z* 215), and $[MH - 116–90]^+$ (*m/z* 243). The loss of 116 Da indicates a hydroxy group on C17[34], which points towards dHy-3 to be drostanolone. This was confirmed by analysing the reference material of drostanolone where the losses of 44 and 144 Da were also observed but were absent in the spectra of the main drostanolone metabolite. Therefore, it can be concluded that the losses of 44 and 144 Da are characteristic of the drostanolone core structure. Furthermore, it is likely that the 144 Da loss originates from the fragmentation of the A ring, but further data collection is needed to substantiate the claim.

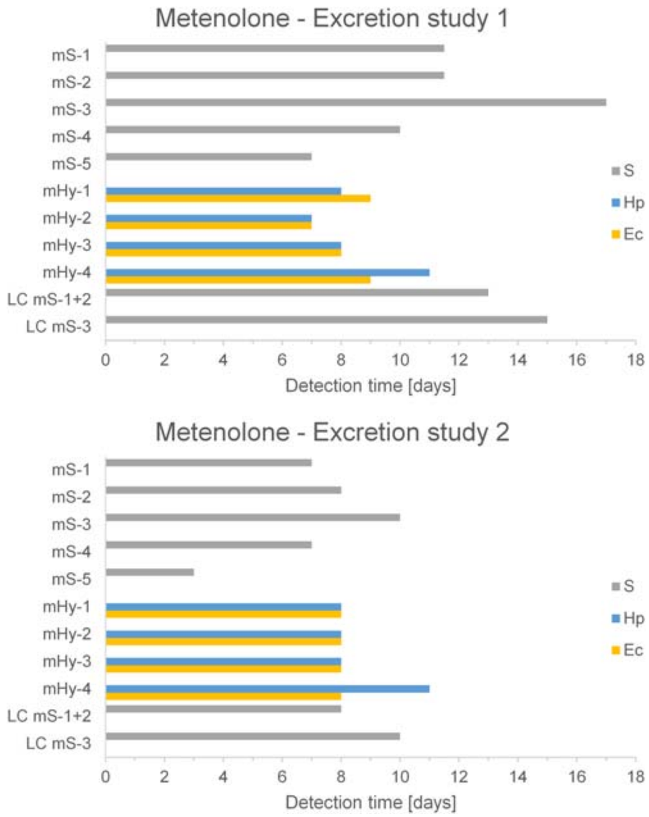


FIGURE 3 Gas chromatography–mass spectrometry detection times of the metabolites of metenolone detected using the three sample preparation methodologies, non-hydrolysed sulfates (S) and hydrolysis with *E. coli* (Ec)-derived β-glucuronidase and *H. pomatia* (Hp)-derived β-glucuronidase/arylsulfatase, and LC–MS detection times of mS-1+2 and mS-3 [Colour figure can be viewed at wileyonlinelibrary.com]

3.3.1.2 | dHy-2

With a mass of *m/z* 361 ($[MH]^+$) in CI [Figure S8 (supporting information)] and *m/z* 360.2843 ($[M]^+$, $C_{23}H_{40}OSi$) in LE-EI-FS, its formula points towards a diol that has lost a single trimethylsilanol (TMSOH) group in the source. This is common for anabolic steroids without a keto-function, and so the precursor is in fact $[MH - 90]^+$.[34] The CI-PIS contained the characteristic ions of steroids without keto-functions (*m/z* 175, 161, 149, 135, 121, etc.). Furthermore, $[MH - 116]^+$ (*m/z* 245) was present, a characteristic loss for hydroxyls attached to C-17[34] indicating that it is the TMSOH group on C3 that is lost. Diols of drostanolone have been reported with various 3α/β versus 17α/β combinations.[22,24,25] As this was not a major metabolite, it falls out of the scope of this study in determining its structure in detail.

3.3.1.3 | dHy-4

The mass of dHy-4 in CI was *m/z* 537 ($[MH]^+$) and *m/z* 536.3532 ($[M]^+$, $C_{29}H_{56}O_3Si_3$) in LE-EI-FS, which is a mass equivalent to drostanolone with an additional hydroxyl group. The CI-PIS spectra [Figure S9 (supporting information)] show extensive fragmentation, but the high mass fragments that are present are losses of TMSOH (90, 180 and 270 Da). This indicates a main drostanolone metabolite core structure as the losses of 116 and/or 144 Da (characteristic losses of the drostanolone core structure) are absent. In the LE-EI-FS spectra [Figure S10 (supporting information)], the ions *m/z* 405.2640 (measured *m/z* 405.2633, $C_{23}H_{41}O_2Si_2$, error 1.63 ppm) and 431.2796 (measured *m/z* 431.2791, $C_{25}H_{43}O_2Si_2$, error 1.18 ppm) were present. The *m/z* 405 ion arises due to the fission of C1/C9 and C2/C3 bonds,[24] indicating that the additional hydroxyl group is on C2. The *m/z* 431 and 405 ions are both present in the work of Liu et al.[25] in a similar metabolite that was determined to have the additional hydroxyl group on C2. Taking all this into account, it is likely that dHy-4 is 2β,3α-dihydroxy-2α-methyl-5α-androstan-17-one, but the synthesis of the reference material is required to confirm the findings.

TABLE 3 GC-CI MRM method for the hydrolysed and non-hydrolysed sulfated metabolites of metenolone

Metabolites	Rt (min)	Precursors [*m/z*]	Product ions [*m/z*] and collision energy (eV)
Hydrolysed			
mHy-1	11.16	447	267(15)–119(15)–105(25)–145(25)
mHy-2	12.26	447	209(25)–187(15)–105(25)–119(15)–277(15)
mHy-3	12.91	533	443(15)–353(25)–295(15)–169(25)–181(25)
mHy-4	13.29	533	207(25)–363(25)–269(25)–181(25)
Non-hydrolysed sulfated			
mS-1	7.41	357	105(32)–161(20)–157(20)–149(20)–133(20)–109(20)
mS-2	7.69	357	119(32)–187(20)–171(13)–161(18)–147(25)–135(20)–133(32)
mS-3	7.98	359	119(22)–173(25)–161(20)–147(25)–145(17)–135(20)–105(32)
mS-4	8.89	357	119(32)–223(20)–209(25)–187(24)–159(27)–147(32)–131(32)
mS-5	10.59	357	109(22)–185(15)–181(32)–143(32)–133(25)–121(25)

Abbreviation: GC-CI MRM, gas chromatography chemical ionization multiple reaction monitoring.

3.3.1.4 | dHy-5

The mass of dHy-5 in CI was *m/z* 535 ($[MH]^+$) and *m/z* 534.3375 ($[M]^+$, $C_{29}H_{54}O_3Si_3$) in LE-EI-FS. When its formula was compared with dHy-4, it lacked two hydrogens. As a metabolite with a double bond has not been reported for drostanolone, it is more likely that dHy-5 originally contained two additional hydroxyl groups, but one is cleaved off in the source as in dHy-2. The CI-PIS [Figure S11 (supporting information)] contains $[MH - 144]^+$ (*m/z* 391), which is a characteristic loss of the drostanolone core structure. Thus, most likely dHy-5 has a drostanolone core structure with two additional hydroxyl groups.

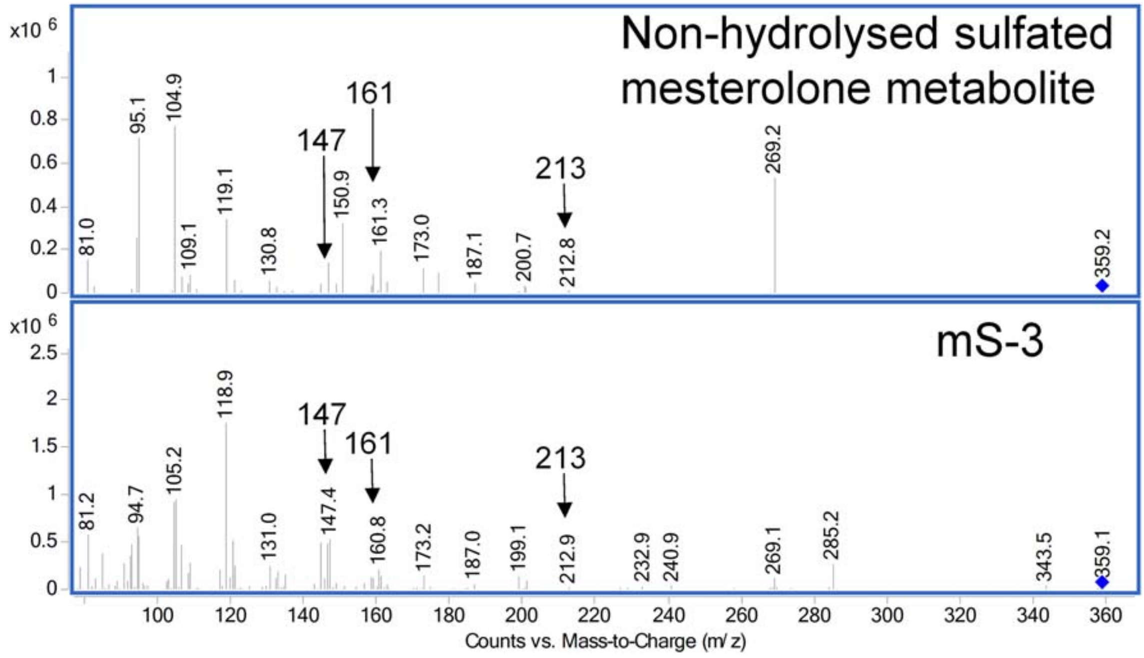


FIGURE 4 Chemical ionization product ion scan of the non-hydrolysed sulfated mesterolone metabolite (15 eV) and the non-hydrolysed sulfated metabolite mS-3 (10 eV) [Colour figure can be viewed at wileyonlinelibrary.com]

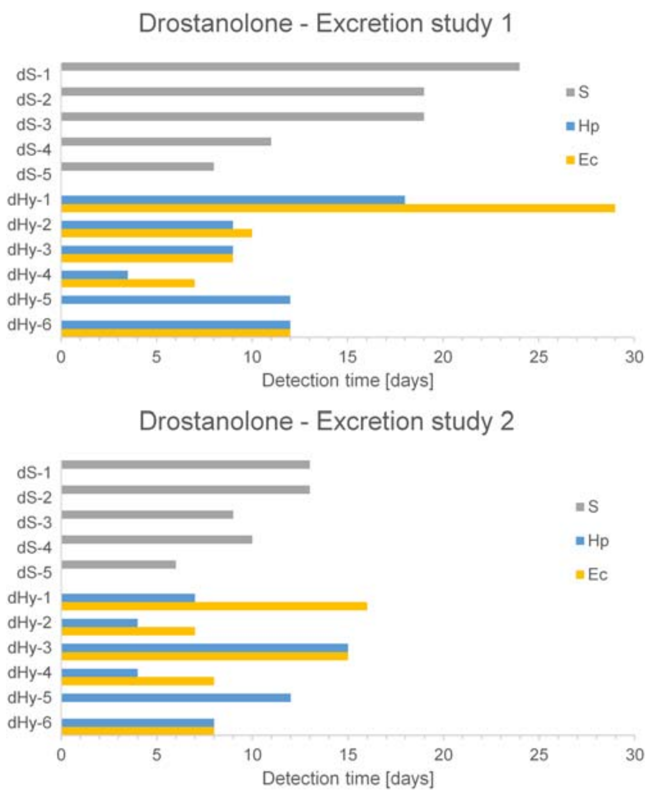


FIGURE 5 Gas chromatography–mass spectrometry detection times of the metabolites of drostanolone detected using the three sample preparation methodologies, non-hydrolysed sulfates (S) and hydrolysis with *E. coli* (Ec)–derived β-glucuronidase and *H. pomatia* (Hp)–derived β-glucuronidase/arylsulfatase [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 4 GC-CI MRM method for the hydrolysed and non-hydrolysed sulfated metabolites of drostanolone

Metabolites	Rt (min)	Precursors [*m/z*]	Product ions [*m/z*] and collision energy (eV)
Hydrolysed			
dHy-1	10.75	449	269(15)–161(30)–145(25)
dHy-2	11.00	361	175(20)–149(25)
dHy-3	12.00	449	305(20)–215(25)
dHy-4	12.43	537	447(10)–357(10)
dHy-5	13.00	535	445(10)–391(30)–355(15)
dHy-6	13.16	625	445(10)–265(30)
Non-hydrolysed sulfated			
dS-1	7.08	359	105(25)–119(25)
dS-2	11.15	447	149(25)–331(15)
dS-3	11.32	449	269(10)–105(25)
dS-4	11.81	447	193(25)–303(20)–207(25)
dS-5	12.34	535	265(20)–117(25)

Abbreviation: GC-CI MRM, gas chromatography chemical ionization multiple reaction monitoring.

3.3.1.5 | dHy-6

The mass of dHy-6 in CI was *m/z* 625 ($[MH]^+$) and *m/z* 624.3876 ($[M]^+$, $C_{32}H_{64}O_4Si_4$) in LE-EI-FS, which corresponded to a mass equivalent to drostanolone with two additional hydroxyl groups. In the CI-PIS spectra [Figure S12 (supporting information)], the losses of four TMSOH groups (*m/z* 535, 445, 355, 265) are evident. The absence of the losses of 144 and 116 Da points towards a main drostanolone metabolite core structure with two additional hydroxyl groups.

3.3.2 | Non-hydrolysed sulfated metabolites

3.3.2.1 | dS-1

The mass of dS-1 in CI was *m/z* 359 ($[MH\text{-}H_2SO_4]^+$) and *m/z* 358.2686 ($[M\text{-}H_2SO_4]^+$, $C_{23}H_{38}OSi$) in LE-EI-FS which corresponded to a mass equivalent to sulfated drostanolone. In CI-PIS (Figure 6), the presence of *m/z* 199 and 213 indicates that the keto group is present on C17,[34] suggesting that dS-1 is the sulfated main drostanolone metabolite. To fully authenticate this, the standard reference material of main drostanolone metabolite was sulfated and its retention time and CI-PIS matched that of dS-1.

3.3.2.2 | dS-2 and dS-4

Both dS-2 and dS-4 had a mass of *m/z* 447 ($[MH - H_2SO_4]^+$) in CI and *m/z* 446.3031 ($[M - H_2SO_4]^+$ $C_{26}H_{46}O_2Si_2$) in LE-EI-FS. This corresponded to a mass equivalent to sulfated drostanolone with an additional hydroxyl group. The CI-PIS of dS-4 [Figure S13 (supporting information)] contains $[MH - 44]^+$ (*m/z* 403) and $[MH - 144]^+$ (*m/z* 303), which are characteristic losses of the drostanolone core structure, whereas for dS-2, these fragments are absent. This along with the fact that dS-4 elutes after dS-2 points towards dS-2 as hydroxylated monosulfated main drostanolone metabolite and dS-4 as hydroxylated monosulfated drostanolone.

3.3.2.3 | dS-3

The mass of dS-3 in CI was *m/z* 449 ($[MH - H_2SO_4]^+$), but its signal in LE-EI-FS was prone to high levels of interference and unfortunately its accurate mass could not be obtained. However, using the information gathered from the measurements in CI, it can be reasoned that dS-3 is a monosulfated diol as the mass difference between dS-2/dS-4 and dS-3 is equal to two hydrogens. The CI-PIS [Figure S14 (supporting information)] shows extensive fragmentation, and the ions characteristic of anabolic steroids without a keto function are present (*m/z* 95, 105 and 119).[34] Furthermore, the absence of a $[MH - 116]^+$ fragment indicates that the sulfate group is on C17, but the synthesis of reference material is required to confirm this.

3.3.2.4 | dS-5

The mass of dS-5 in CI was *m/z* 535 ($[MH - H_2SO_4]^+$) and *m/z* 534.3375 ($[M - H_2SO_4]^+$, $C_{29}H_{54}O_3Si_3$) in LE-EI-FS, which is a mass equivalent of a monosulfated structure with two additional hydroxyl groups. The most abundant fragments in CI-PIS [Figure S15 (supporting information)] are losses of TMSOH groups (*m/z* 445, 355 and 265) and the losses of 116 or 144 Da are absent. This points towards a monosulfated main drostanolone metabolite structure that contains two additional hydroxyl groups.

3.4 | Considerations for application in anti-doping control

The detection time graph for metenolone for both excretion studies is presented in Figure 3. The detection times range from 3 to 17 days with the non-hydrolysed sulfated metabolite mS-3 providing the longest detection time of 17 days. For metenolone, focusing on non-hydrolysed sulfated metabolites substantially increased the detection window. In the first excretion study, this increased the detection time by nearly a factor of 2 in comparison with the currently monitored markers for metenolone (i.e. hydrolysed glucuronides) in a routine doping control initial testing procedure. In the second excretion study, it prolonged the detection window by 25%. In addition, both excretion studies were analysed on LC–MS/MS, focusing on the two longest detectable non-hydrolysed sulfated metabolites on GC (Figure 3). The detection times on GC either equal, as in the case of excretion study 2, or surpass the detection times on LC. This further suggests the potential of using GC–MS/MS to detect non-hydrolysed sulfated steroids in a routine initial testing procedure.

The results of the drostanolone excretion studies are presented in Figure 5 with detection times up to 29 days. The metabolite providing the longest detection time in both excretion studies is dHy-1 with hydrolysis using *E. coli*–derived β-glucuronidase. The non-hydrolysed sulfate providing the longest detection time is dS-1, the sulfated analogue of dHy-1, which differs from the one reported by Liu et al.[25] In the case of drostanolone, non-hydrolysed sulfated metabolites turn out to provide no added value over the currently monitored target compounds (i.e. hydrolysed glucuronides).

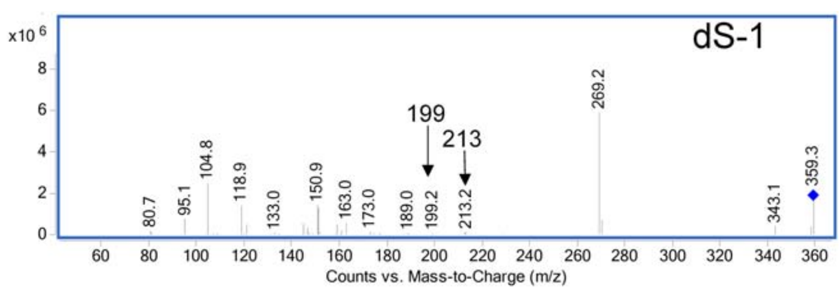


FIGURE 6 Chemical ionization product ion scan of the non-hydrolysed metabolite dS-1 (10 eV) [Colour figure can be viewed at wileyonlinelibrary.com]

4 | CONCLUSIONS

Through two excretion studies, a search for new metabolites of metenolone and drostanolone was undertaken with the analysis carried out on GC–CI–MS/MS rather than LC–MS. The detection protocol was applied to three separate sample preparation methodologies (hydrolysis with *E. coli*–derived β-glucuronidase, *H. pomatia*–derived β-glucuronidase/arylsulfatase and non-hydrolysed sulfated steroids).

For metenolone, a non-hydrolysed sulfated metabolite, identified as 1β-methyl-5α-androstan-17-one-3ζ-sulfate, proved to have superior detection time over the hydrolysed metabolites where the detection window was almost double that of the hydrolysed metabolite. A comparison of the two applied hydrolysis procedures points out that *H. pomatia*–derived β-glucuronidase/arylsulfatase outperformed *E. coli*–derived β-glucuronidase which is generally used as a sample procedure. However, regulations only permit the use of *E. coli*–derived β-glucuronidase in the evaluation of the endogenous steroid profile,[37] and thus most anti-doping laboratories continue their use of *E.coli*–derived β-glucuronidase.

For drostanolone, hydrolysis with *E. coli*–derived β-glucuronidase of the main drostanolone metabolite (3α-hydroxy-2α-methyl-5α-androstan-17-one) proved to have the longest detection window in both excretion studies with detection times up to 29 days. A new potential long-term sulfated metabolite was identified as 2α-methyl-5α-androstan-17-one-3α-sulfate, a sulfated analogue of the main drostanolone metabolite. Furthermore, a greater degree of hydroxylation of the metabolites of drostanolone was detected than has been previously reported. This supports the theory that the detection protocol applied in this study provides a relatively easy way to investigate the potential of less likely metabolites.

ACKNOWLEDGEMENT

This project was carried out with the support of the Partnership for Clean Competition (Grant Number: 2019R2000409G).

ORCID

Aðalheiður Dóra Albertsdóttir https://orcid.org/0000-0003-2748-6197

Michael Polet https://orcid.org/0000-0003-0801-5891

REFERENCES

1. World Anti-Doping Agency. 2018 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/2018_testing_figures_report.pdf. Published 2018. Accessed January 2, 2020.
2. World Anti-Doping Agency. 2017 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/2017_anti-doping_testing_figures_en_0.pdf. Published 2017. Accessed June 20, 2019.
3. World Anti-Doping Agency. 2016 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/2016_anti-doping_testing_figures.pdf. Published 2016. Accessed May 14, 2019.
4. World Anti-Doping Agency. 2015 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/2015_wada_anti-doping_testing_figures_report_0.pdf. Published 2015. Accessed July 3, 2019.
5. World Anti-Doping Agency. 2014 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/wada_2014_anti-doping-testing-figures_full-report_en.pdf. Published 2014. Accessed June 20, 2019.
6. World Anti-Doping Agency. 2013 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/wada_2013_anti-doping_testing_figures_report_en.pdf. Published 2013. Accessed April 3, 2020.
7. World Anti-Doping Agency. 2012 Anti-Doping Testing Figures. World Anti-Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/WADA-2012-Anti-Doping-Testing-Figures-Report-EN.pdf. Published 2012. Accessed April 3, 2020.
8. Thevis M, Kuuranne T, Geyer H. Annual banned-substance review: analytical approaches in human sports drug testing. *Drug Test Anal*. 2019;11(1):8-26. https://doi.org/10.1002/dta.2549
9. Polet M, Van Gansbeke W, Geldof L, Deventer K, Van Eenoo P. Identification and characterization of novel long-term metabolites of oxymesterone and mesterolone in human urine by application of selected reaction monitoring GC-CI-MS/MS. *Drug Test Anal*. 2017;9(11–12):1673-1684. https://doi.org/10.1002/dta.2183
10. Gómez C, Pozo OJ, Garrostas L, Segura J, Ventura R. A new sulphate metabolite as a long-term marker of metandienone misuse. *Steroids*. 2013;78(12):1245-1253. https://doi.org/10.1016/j.steroids.2013.09.005
11. Gómez C, Pozo OJ, Marcos J, Segura J, Ventura R. Alternative long-term markers for the detection of methyltestosterone misuse. *Steroids*. 2013;78(1):44-52. https://doi.org/10.1016/j.steroids.2012.10.008
12. Balcells G, Pozo OJ, Garrostas L, et al. Detection and characterization of clostebol sulfate metabolites in Caucasian population. *J Chromatogr B*. 2016;1022:54-63. https://doi.org/10.1016/j.jchromb.2016.03.028
13. Van Hoof N, Courtheyn D, Gillis W, et al. Metabolism of Methenolone acetate in a veal calf. *Vet Res Commun*. 2007;31(3):259-272. https://doi.org/10.1007/s11259-006-3432-z
14. Okamoto S, Sonoda S, Tanino G, Tomida K, Okazaki H, Kondo I. Change in thigh muscle cross-sectional area through Administration of an Anabolic Steroid during routine stroke rehabilitation in hemiplegic patients. *Am J Phys Med Rehabil*. 2011;90(2):106-111. https://doi.org/10.1097/PHM.0b013e31820172bf
15. Shahidi NT. A review of the chemistry, biological action, and clinical applications of anabolic-androgenic steroids. *Clin Ther*. 2001;23(9):1355-1390. https://doi.org/10.1016/S0149-2918(01)80114-4
16. Björkhem I, Ek H. Detection and quantitation of 3α-hydroxy-1-methylen-5α-androstan-17-one, the major urinary metabolite of methenolone acetate (Primobolan®) by isotope dilution—mass spectrometry. *J Steroid Biochem*. 1983;18(4):481-487. https://doi.org/10.1016/0022-4731(83)90069-9
17. Fragkaki AG, Angelis YS, Kiousi P, Georgakopoulos CG, Lyris E. Comparison of sulfo-conjugated and gluco-conjugated urinary metabolites for detection of methenolone misuse in doping control by LC-HRMS, GC-MS and GC-HRMS. *J Mass Spectrom*. 2015;50(5):740-748. https://doi.org/10.1002/jms.3583
18. He G, Yang S, Lu J, Xu Y. New long term metabolite in human urine for metenolone misuse by liquid chromatography quadrupole time-of-flight mass spectrometry. *Steroids*. 2016;105:1-11. https://doi.org/10.1016/j.steroids.2015.10.015
19. Gerace E, Salomone A, Abbadessa G, Racca S, Vincenti M. Rapid determination of anti-estrogens by gas chromatography/mass spectrometry in urine: method validation and application to real samples. *J Pharm Anal*. 2012;2(1):1-11. https://doi.org/10.1016/j.jpha.2011.09.011

20. Heney NM. Treatment of advanced breast carcinoma with Drostanolone propionate. *Bristol Med Chir J*. 1970;85(3):75-76.
21. Choudhary M.I., Siddiqui M, Atia-tul-Wahab, et al. Bio-Catalytic Structural Transformation of Anti-cancer Steroid, Drostanolone Enanthate with Cephalosporium aphidicola and Fusarium lini, and Cytotoxic Potential Evaluation of Its Metabolites against Certain Cancer Cell Lines. *Front. Pharmacol*. 2017;8:900. https://doi.org/10.3389/fphar.2017.00900
22. Gauthier J, Goudreault D, Poirier D, Ayotte C. Identification of drostanolone and 17-methyldrostanolone metabolites produced by cryopreserved human hepatocytes. *Steroids*. 2009;74(3):306-314. https://doi.org/10.1016/j.steroids.2008.11.002
23. Templeton JF. Ryung-soon song Kim. Metabolism of 17β-hydroxy-2α-methyl-5α-androstan-3-one in the rabbit. *Steroids*. 1977;29(3):371-381. https://doi.org/10.1016/0039-128X(77)90006-X
24. De Boer D, de Jong EG, Maes RAA, van Rossum JM. The methyl-5α-dihydrotestosterones mesterolone and drostanolone; gas chromatographic/mass spectrometric characterization of the urinary metabolites. *J Steroid Biochem Mol Biol*. 1992;42(3):411-419. https://doi.org/10.1016/0960-0760(92)90146-A
25. Liu Y, Lu J, Yang S, Zhang Q, Xu Y. New drostanolone metabolites in human urine by liquid chromatography time-of-flight tandem mass spectrometry and their application for doping control. *Steroids*. 2016;108:61-67. https://doi.org/10.1016/j.steroids.2016.01.013
26. Polet M, Van Gansbeke W, Van Eenoo P. Development and validation of an open screening method for doping substances in urine by gas chromatography quadrupole time-of-flight mass spectrometry. *Anal Chim Acta*. 2018;1042:52-59. https://doi.org/10.1016/j.aca.2018.08.050
27. Pereira HMG, Sardela VF, Padilha MC, et al. Doping control analysis at the Rio 2016 Olympic and Paralympic games. *Drug Test Anal*. 2017;9(11–12):1658-1672. https://doi.org/10.1002/dta.2329
28. Jiménez Girón A, Deventer K, Roels K, Van Eenoo P. Development and validation of an open screening method for diuretics, stimulants and selected compounds in human urine by UHPLC–HRMS for doping control. *Anal Chim Acta*. 2012;721:137-146. https://doi.org/10.1016/j.aca.2012.02.002
29. Van Gansbeke W, Polet M, Hooghe F, Devos C, Van Eenoo P. Improved sensitivity by use of gas chromatography—positive chemical ionization triple quadrupole mass spectrometry for the analysis of drug related substances. *J Chromatogr B*. 2015;1001:221-240. https://doi.org/10.1016/j.jchromb.2015.07.052
30. Polet M, Van Gansbeke W, Albertsdóttir AD, Coppieters G, Deventer K, Van Eenoo P. Gas chromatography—mass spectrometry analysis of non-hydrolyzed sulfated steroids by degradation product formation. *Drug Test Anal*. 2019;11(11-12):1-10. https://doi.org/10.1002/dta.2606
31. Cha E, Kim S, Kim HJ, et al. Sensitivity of GC-EI/MS, GC-EI/MS/MS, LC-ESI/MS/MS, LC-ag+CIS/MS/MS, and GC-ESI/MS/MS for analysis of anabolic steroids in doping control. *Drug Test Anal*. 2015;7(11–12):1040-1049. https://doi.org/10.1002/dta.1906
32. Waller CC, McLeod MD. A simple method for the small scale synthesis and solid-phase extraction purification of steroid sulfates. *Steroids*. 2014;92:74-80. https://doi.org/10.1016/j.steroids.2014.09.006
33. Polet M, Van Gansbeke W, Van Eenoo P, Deventer K. Efficient approach for the detection and identification of new androgenic metabolites by applying SRM GC-CI-MS/MS: a methandienone case study. *J Mass Spectrom*. 2016;51(7):524-534. https://doi.org/10.1002/jms.3781
34. Polet M, Van Gansbeke W, Van Eenoo P, Deventer K. Gas chromatography/chemical ionization triple quadrupole mass spectrometry analysis of anabolic steroids: ionization and collision-induced dissociation behavior. *Rapid Commun Mass Spectrom*. 2016;30(4):511-522. https://doi.org/10.1002/rcm.7472
35. Fragkaki AG, Angelis YS, Tsantili-Kakoulidou A, Koupparis M, Georgakopoulos C. Schemes of metabolic patterns of anabolic androgenic steroids for the estimation of metabolites of designer steroids in human urine. *J Steroid Biochem Mol Biol*. 2009;115(1):44-61. https://doi.org/10.1016/j.jsbmb.2009.02.016
36. Goudreault D, Massé R. Studies on anabolic steroids—4. Identification of new urinary metabolites of methenolone acetate (primobolan®) in human by gas chromatography/mass spectrometry. *J Steroid Biochem Mol Biol*. 1990;37(1):137-154. https://doi.org/10.1016/0960-0760(90)90383-V
37. WADA. TD2018EAAS. World Anti-Doping Agency. https://www.wada-ama.org/en/resources/science-medicine/td2018eaas-0. Published 2018. Accessed February 18, 2020.

How to cite this article: Albertsdóttir AD, Van Gansbeke W, Coppieters G, Balgimbekova K, Van Eenoo P, Polet M. Searching for new long-term urinary metabolites of metenolone and drostanolone using gas chromatography–mass spectrometry with a focus on non-hydrolysed sulfates. *Drug Test Anal*. 2020;12:1041–1053. https://doi.org/10.1002/dta.2818